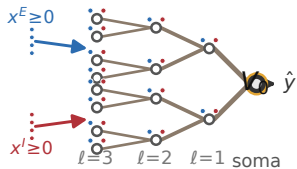
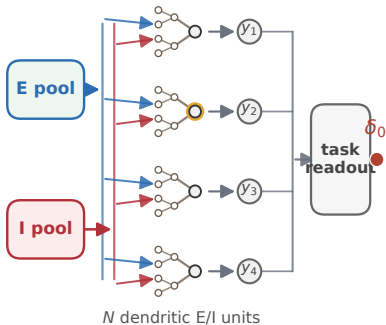


A Single dendritic E/I unit

● E ● I

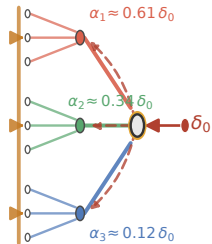


B Network layer



C Credit assignment and broadcast

rank-1
 e_n shared



Broadcast modes

Rank-1
shared scalar →

Rank-K
 K random channels →

Path
branch-structured →

Oracle
 $\alpha_n \delta_0$ →

rank-1 $\approx$ exact only when α_n is compressible

$$\partial L / \partial g_i = x_i R_n^{\text{tot}} (E_i - \delta_{nn}) \Delta g_i \propto x_i R_n^{\text{tot}} (E_i - \delta_{nn}) e_n \approx \delta_n$$